

Define Problem Statement

Credit Card Approval Prediction System | Brainstorming & Ideation Phase

Background

Banks and financial institutions receive thousands of credit card applications every day. Evaluating each one manually is time-consuming, inconsistent, and prone to human bias. A key challenge is identifying applicants who are likely to default or make late payments — these are high-risk profiles that should be rejected to protect the lender.

Problem Statement

Given an applicant's demographic information (gender, age, family status, education, housing type, occupation) and financial profile (income, car/realty ownership, employment history), and their past credit behaviour (payment STATUS from credit records), there is no lightweight, interactive tool for a loan officer or analyst to instantly predict whether a credit card application should be **Approved** or **Rejected**.

Root Cause

- Manual review is slow and inconsistent across officers.
- Credit history data (STATUS codes 1-5 = late/default) is scattered across a separate credit record table and not automatically merged with application data.
- No decision-support tool exists that combines both application demographics and credit behaviour into a single prediction.

Proposed Solution

Build a **Credit Card Approval Prediction System** — a Flask web application backed by a Random Forest classifier trained on merged UNDP-style credit data (438,557 application records + 1,048,575 credit records). The system accepts 17 applicant features through a polished dark-theme form and instantly outputs an **Approved** or **Rejected** decision with visual feedback.

Scope

In Scope	Out of Scope
Binary classification: Approved (0) / Rejected (1)	Probability scores or credit limit recommendations
17-feature applicant input via web form	Real-time bank API integration
Random Forest model (best performer after comparison)	Deep learning / neural network approaches
Dark-theme Flask UI with Approved/Rejected result page	Multi-language support
Render deployment via Gunicorn	Mobile native app